

# MTL106: Probability and Stochastic Processes

## Lecture 1

January 2, 2026

# The Need of Probability as a Mathematical Tool

In our day to day life, we come across a lot of uncertainties. Some examples are provided below:

- Predicting quarterly revenue of an organization.
- Number of customers likely to churn before next quarter.
- Demand for a product at a future time period.
- Number of views for an Instagram reel or a YouTube video.
- Outcome of the cricket matches between India and Australia.

# The Need of Probability as a Mathematical Tool (Contd.)

- It is essential to develop a mathematical tool that can be used to predict and measure such uncertainties.
- **Probability** is a fundamental mathematical tool to understand and model the uncertainty associated with an event.

# Random Experiment

A **Random Experiment** is an experiment in which:

- All outcomes of the experiment are known in advance.
- Any performance of the experiment results in an outcome that is not known in advance.
- The experiment can be repeated under identical conditions.

In probability theory we study this uncertainty of a random experiment.

# Sample Space

The collection of all possible outcomes of a random experiment is called the **Sample Space**. Usually a sample space is denoted by  $\Omega$ .

- Outcome of a cricket match:  $\Omega = \{\text{Win, Lose, Draw}\}$ .
- Customer Churn:  $\Omega = \{\text{Churn, No Churn}\}$ .
- Analyzing YouTube views:  $\Omega = \{0, 1, 2, \dots, 1000, \dots, 10000, \dots\}$ .
- Lifetime ( $T$ ) of a new product:  $\Omega = \{T \mid 0 < T < \infty\}$ .

An **Event** ' $E$ ' is a subset of the sample space  $\Omega$  ( $E \subseteq \Omega$ ) and we calculate probability with respect to the event  $E$ . So if the outcome of a random experiment is a member of the set  $E$ , one can say that the event  $E$  has occurred.

- Customer churn ( $C$ ) from a customer base of 100 customers:

$$\Omega = \{C \mid 0 \leq C \in \mathbb{Z} \leq 100\}.$$

$$E_1 = \{C \mid C < 20\}.$$

$$E_2 = \{C \mid C > 40\}.$$

$$E_3 = \{C \mid 20 < C < 40\}.$$

- Outcome of a cricket match:

$$\Omega = \{\text{Win}, \text{Lose}, \text{Draw}\}.$$

$$E_1 = \{\text{Win}\}.$$

$$E_2 = \{\text{Lose}\}.$$

$$E_3 = \{\text{Draw}\}.$$

# Event (Contd.)

**Remark 1.** Two events  $E_1$  and  $E_2$  are said to be **Mutually Exclusive**, if they cannot occur simultaneously, i.e.,  $E_1 \cap E_2 = \emptyset$ , the empty set.

- If two competing companies are bidding for the same contract, only one can win. If company  $A$  wins, Company  $B$  cannot, and vice versa.
- The outcome of a surgery could be either successful or unsuccessful. These outcomes are mutually exclusive.

**Remark 2.** **Equally Likely** events are those that have the same probability of occurring.

- Lottery for holiday leave.
- Random distribution of promotional coupons.
- Random customer survey selection.

# Assignment of Probabilities

It might be possible that in a random experiment some events are more likely to occur than the others. See following examples:

- In a match between a top-ranked team and a lower-ranked team, the higher-ranked team is more likely to win.
- A person with a high credit score is more likely to get a loan approved than someone with a low credit score.



# Assignment of Probabilities (Contd.)

- It is desirable to measure the likelihoods of the occurrences of various events.
- Probability of an event is a quantitative measure of uncertainty with which that event occurs.
- We assign a real number  $P(E) \in [0, 1]$  to each event.
- We must have  $P(\Omega) = 1$ .

# Classical Approach

- The **Classical Approach** of probability assignment is used when there are finite number of equally likely outcomes for some random experiment.
- Suppose  $\Omega = \{\omega_1, \omega_2, \dots, \omega_k\}$  represents a finite sample space with  $k (\in \mathbb{N})$  possible outcomes.
- Let  $E \subseteq \Omega$  is an event and  $n(E)$  denotes the number of elements in  $E$ .
- Let  $\omega \in \Omega$  is favorable to an event  $E$ , i.e.,  $\omega \in E$ .
- According to the classical approach, the probability of the occurrence of an event  $E$  is the ratio of the *number of favorable outcomes to  $E$*  and *the total number of outcomes*, i.e.,

$$P(E) = \frac{n(E)}{n(\Omega)} = \frac{n(E)}{k}.$$

# Classical Approach (Contd.)

For an event  $E$ , the probabilities assigned through the classical approach satisfy the following conditions:

- 1  $P(E) \geq 0$ .
- 2  $P\left(\bigcup_{i=1}^k E_i\right) = \sum_{i=1}^k P(E_i)$ , where  $E_1, E_2, \dots, E_k$  are mutually exclusive events ( $E_i \cap E_j = \emptyset$ ,  $i, j \in \{1, 2, \dots, k\}$ ,  $i \neq j$ ).
- 3  $P(\Omega) = 1$ .

**Remark 3.** The classical approach of probability assignment is confined to those random experiments only which have a finite number of equally likely outcomes. This makes the use of the classical approach very restricted.

# Relative Frequency Approach

- Suppose we repeat a random experiment under identical conditions such that the trials are independent.
- Let  $f_N(E)$  denotes the frequency of an event  $E$  in  $N$  independent trials.
- Let  $r_N(E) = \frac{f_N(E)}{N}$  represents the corresponding relative frequency.
- Then according to the **Relative Frequency Approach**, probability of the event  $E$  is given by

$$P(E) = \lim_{N \rightarrow \infty} r_N(E) = \frac{f_N(E)}{N}.$$

- Note that the experiment is repeated a large number ( $N$ ) of times and the approximation  $P(E) \approx r_N(E)$  is used for assigning probability to the event  $E$ .

# Relative Frequency Approach (Contd.)

For an event  $E$ , the probabilities assigned through the relative frequency approach also satisfy the following conditions:

- ①  $P(E) \geq 0$ .
- ②  $P\left(\bigcup_{i=1}^k E_i\right) = \sum_{i=1}^k P(E_i)$ , where  $E_1, E_2, \dots, E_k$  are mutually exclusive events ( $E_i \cap E_j = \emptyset$ ,  $i, j \in \{1, 2, \dots, k\}$ ,  $i \neq j$ ).
- ③  $P(\Omega) = 1$ .

**Remark 4.** Since the relative frequency approach is based on an approximation, it might not be precise enough. Also due to various logistical constraints, it is not always feasible to repeat the random experiment a large number of times. These are the limitations of the relative frequency approach.